2021

因子研究 03：191 因子算法及其计算说明

量化数据

深圳天软科技

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1 更新日志

1.1 更新日志

更新日志	更新说明
2020-03-11	创建文档
2020-11-13	添加“参考文档”
2021-11-26	平台因子整合，文档调整

1.2 摘要

1) 提供了 191 因子的函数开发算法、提取范例以及常用的底层算法说明。

2) 提供了 191 因子底层缓存机制的处理和特殊因子公式的调整方案。

2 Alpha191 因子介绍

2017 年 6 月，国泰君安证券研究者结合 101alpha 因子在研报中推出了<<**数量化专题：基于短周期价量特征的多因子选股体系**>>。其中包含 191 个因子表达式。我们把这些公式编写成了天软函数，供大家使用。

本文主要介绍两部分：

1) 191 因子公式常使用到的算法。

2) 如何快速从天软提取对应的因子值。

至于 191Alpaha 因子中哪些因子具有显著性或者说能否适合用在 A 股市场等问题，需要大家去探究，本文致力于大家在探究 191Alpaha 因子时提供一些中间环节的帮助。

3 191 因子公式计算函数

3.1 函数开发标准

191 因子函数在天软中的路径为：公用函数\\扩展函数\\.Net 扩展函数\\技术分析\\191 因子文件夹下，所有因子的函数名称以 Tsgt_开头作为前缀。

191 因子计算时， 大部分的算法不仅会计算和其他股票对比的排名， 也会用到个股自身历史数据，这就引发出两个问题：

1) 如果要进行全市场排名时，股票池的范围如何定义，是否要考虑剔除停牌等因素？

2) 取历史数据时，遇到停牌如何处理？

针对上述问题， 我们给出了 191 因子 函数规则：

- 计算全市场排名时， 用来排名的股票池范围需要用户设置，详情看范例 3.1.1；
- 对于停牌个股， 停牌期间不予计算， 一律返回 NAN 值；
- 对于正常交易的个股， 使用到历史数据时，会考虑时间对齐，一律使用市场交易日序列；
- 对于正常交易的个股，使用市场交易日序列取历史数据，会遇到历史停牌的情况，101Alpaha 因子开发的标准是，删掉停牌时的数据，若因删掉数据而计算不出最终的因子值时，一律返回 NAN 值。

3.1.1 计算 191 因子范例

```
EndT:=20190402T;
StockArr:=GetBKByDate('SH000300',EndT);

SetSysParam(pn_StockArr(),StockArr); //注意设置 pn_StockArr()
```

```
SetSysParam(PN_Cycle(),cy_day());
SetSysParam(PN_Rate(),1);
Setsysparam(pn_rateday(),endt);
SetSysParam(PN_Date(),EndT);
r:=array();
n:=0;
for nI:=0 to length(StockArr)-1 do
begin
   StockId:=StockArr[nI];
   echo StockId;
   SetSysParam(PN_Stock(),StockId);
   r[n]['代码']:=StockId;
   r[n]['名称']:=StockName(StockId);
   r[n]['Alph002']:=Tsgt_Alpha_002();
   r[n]['Alph003']:=Tsgt_Alpha_003();
   //其他的模型类似的调用方法，如： Tsgt_Alpha_xxx，其中 xxx 为对应的序号
   //r[n]['Alph00N']:=Tsgt_Alpha_xxx();
   n++;
end
return r;
```

3.2 常用算法

编号	191 因子函数	天软函数	备注
01	REGBETA(A, B, n)	SlopeofNday(Exp1,Exp2,N,CacheStr)	新增
02	REGRESI(A, B, n)	interceptofNday(Exp1,Exp2,N,CacheStr)	新增
03	SMA(A, n, m)	SMaofNday(Exp,N,M,CacheAstr)	新增
04	SUMIF(A, n, condition)	SumifofNday(Exp,N,Condition,CacheAstr)	新增
05	SEQUENCE(n)	GT_Sequence(N)	新增
06	RANK(A)	Percentrankoftheday(Exp,Cachestr)	参考 101 因子算法说明
07	STD(A, n)	StdofNday(Exp,N,CacheStr)	
08	CORR(A, B, n)	CorrelOfNDay(Exp1,Exp2,N,CacheStr)	
09	DELTA(A, n)	DeltaOfNDay(Exp,N)	
10	SUM(A, n)	SumofNday(Exp,N,CacheAstr)	
11	MEAN(A, n)	AvgofNday(Exp,N,CacheAstr)	
12	TSRANK(A, n)	percentrankofNday(Exp,N,Cachestr)	
13	COVIANCE (A, B, n)	CovarOfNDay(Exp1,Exp2,N,CacheStr)	
14	DELAY(A, n)	refvalue(Exp,N)	
15	TSMIN(A, n)	LowestOfNDay(Exp,N,CacheStr)	

16	TSMAX(A, n)	HighestOfNDay(Exp,N,CacheStr)	
17	PROD(A, n)	ProductOfNDay(Exp,N,CacheStr)	
18	COUNT(condition, n)	OccuredOfNDay(Exp,N)	
19	WMA(A, n)	WmaofNday(Exp,N,CacheAstr)	
20	DECAYLINEAR(A, d)	DecayAvgOfNDay(Exp,N,CacheAstr)	
21	HIGHDAY(A, n)	HHVBarsOfNDay(Exp,N,CacheStr)	
22	LOWDAY(A, n)	LLVBarsOfNDay(Exp,N,CacheStr);	

3.2.1 REGBETA(A, B, n)

A,B 序列 N 日的回归系数。这里的 N 指的市场交易日，非交易日数据剔除处理，若区间内均非交易日或当日非个股交易日时返回 NAN。在 TS 对应的函数为 SlopeofNday(Exp1,Exp2,N,CacheStr)，函数说明参照下节。

3.2.1.1 SlopeofNday(Exp1,Exp2,N,CacheStr)

- 定义：SlopeofNday(Exp1,Exp2,N,CacheStr)
- 说明：两个表达式 N 日的回归系数。实数。
- 参数：

  Exp1：表达式 1

  Exp2：表达式 2

  N：市场 N 个交易日

  CacheAstr：缓存串
- 返回：N 日回归系数。

3.2.2 REGRESI(A, B, n)

A,B 序列 N 日的截距。这里的 N 指的市场交易日，非交易日数据剔除处理，若区间内非交易日或当日非个股交易日时返回 NAN。在 TS 对应的函数为 interceptofNday(Exp1,Exp2,N,CacheStr)，函数说明参照下节。

3.2.2.1 interceptofNday(Exp1,Exp2,N,CacheStr)

- 定义：interceptofNday(Exp1,Exp2,N,CacheStr)
- 说明：两个表达式 N 日的截距。

- 参数：

  Exp1：表达式 1

  Exp2：表达式 2

  N：市场 N 个交易日

  CacheAstr：缓存串

- 返回：截距。

3.2.3 SMA(A, n, m)

A 序列 N 日平滑移动平均值。当个股当日停牌时返回 NAN。在天软中对应的函数为 SMaofNday(Exp,N,M,CacheAstr)，函数说明参考下节。

3.2.3.1 SMaofNday(Exp,N,M,CacheAstr)

- 定义：SMaofNday (Exp1, N,M,CacheStr)
- 说明：表达式 N 日平滑移动平均。
- 参数：

  Exp：表达式

  N：N 日

  M：权重，整数

  CacheAstr：缓存串

- 返回：平滑移动平均。

3.2.4 SUMIF(A, n, condition)

A 序列市场 N 日条件求和。当个股当日停牌时返回 NAN。在天软中对应的函数为 SumifofNday(Exp,N,Condition,CacheAstr)，函数说明参考下节。

3.2.4.1 SumifofNday(Exp,N,Condition,CacheAstr)

- 定义：SumifofNday(Exp,N,Condition,CacheAstr)
- 说明：N 日条件求和
- 参数：

  Exp：表达式

  N：市场 N 日

  Condition：条件表达式

CacheAstr：缓存串

- 返回：条件求和。

3.2.5 SEQUENCE(n)

生成 1~N 的等差序列。在天软中对应的函数为 GT_Sequence(N)，函数说明参考下节。

3.2.5.1 GT_Sequence(N)

- 定义：GT_Sequence(N)
- 说明：生成 1~N 的等差序列
- 参数：

  N：序列长度

- 返回：生成 1~N 的等差序列。

4 计算函数的缓存机制

191 因子中，很多算法比较复杂，而且还用到了全市场的数据，举个简单的例子，计算一只股票的因子值的排名，已经算好了，如果计算第二只股票时，又要计算全市场排名，是一件开销很大的事情，因此 191aphla 因子函数封装时使用的了缓存机制，启动缓存机制后，避免了重复计算，加快了速度。

比较常见的就是算法函数中的最后一个参数总是带有 CacheStr， 这个参数起到一个缓存串的作用，而且是唯一的串， 一般这个参数能作为唯一串应该是由因子名称+日期+算法合成，如果是单个因子中需要缓存多个中间变量，则需要加入序号即可。 如下所示：

```
Function Tsgt_Alpha_036();
Begin
    {
    RANK(SUM(CORR(RANK(VOLUME), RANK(VWAP)), 6), 2)
    }
    s:=functionname();
    s1:=s+"01";
    s2:=s+"02";
    s3:=s+"03";
    s4:=s+"04";
    if not IsStockTradeDay(sp_time(1)) then return nan
    else
    begin
       index:=@CorrelOfNDay(@PercentRankOftheDay(@Vol(),s),
                            @PercentRankOftheDay(@StockPjCj3(),s1),
                            6,s2);
       return PercentRankOftheDay(@SumOfNDay(index,2,s3),s4);
    end
End;
```

Tsgt_Alpha_036()函数就缓存了 5 个全市场中间变量，因此同一个因子缓存多个中间变量时需要区分，避免缓存混乱。即使是带这里，从时间维度的缓存还是没有解决，还得需要加入日期+以及算法函数名称等（不过多赘述）。 其他算法的缓存机制也是类似。 而在计算结束后会释放掉所有缓存。

5 因子调整

由于研报中部分因子异常情况，需要做额外处理。常见的异常包含如下：

- 公式异常：

  1) 因子公式中存在公式错误无法正确返回结果的情况，处理方法直接处理为 0（因子废弃）

  2) 因子公式算法异常但可以正常返回结果，处理方法为调整对应算法处理

- 变量调整

  主要指的 Fama French 三因子，其中：

  HML：价值-成长（800 价值指数-800 成长指数）

  MKE：中证 800 收益率

  SMB：调整为小盘收益率-大盘收益率的差值（中证 500-中证 100）

- 结果异常

  主要指的因子的返回结果。由于数据入库的原因，部分数据过小导致无法正常入库，因此需要做额外调整。

调整的因子包含如下：

因子	异常原因	调整情况
Alpha030	变量调整	Fama French 三因子调整
Alpha144	结果异常	结果集*1E8
Alpha009	结果异常	结果集*1E7
Alpha068	结果异常	结果集*1E7
Alpha054	公式异常	Std 添加参数 10
Alpha021	公式异常	Slope 添加参数 6
Alpha127	公式异常	Mean 添加参数 12,Max 改为 TSmax
Alpha146	公式异常	SMA 添加参数 2
Alpha165	公式异常	处理返回 0
Alpha166	公式异常	处理返回 0
Alpha183	公式异常	处理返回 0
Alpha143	公式正常但递推	处理为统计 252 个交易日
Alpha119	公式正常但逻辑不对	将 Min 函数调整为 TSMin
Alpha007	公式正常但逻辑不对	将 Min,Max 函数调整为 TSMin,TSMax
Alpha010	公式正常但逻辑不对	将 Max 函数调整为 TSMax
Alpha017	公式正常但逻辑不对	将 Max 函数调整为 TSMax
Alpha028	公式正常但逻辑不对	将 Max 函数调整为 TSMax
Alpha041	公式正常但逻辑不对	将 Max 函数调整为 TSMax

Alpha091	公式正常但逻辑不对	将 Max 函数调整为 TSMax
Alpha162	公式正常但逻辑不对	将 Min,Max 函数调整为 TSMin,TSMax
Alpha108	公式正常但逻辑不对	将 Min,函数调整为 TSMin
Alpha154	公式正常但逻辑不对	将 Min,函数调整为 TSMin
Alpha121	公式正常但逻辑不对	将 Min,函数调整为 TSMin
Alpha157	公式正常但逻辑不对	将 Min,函数调整为 TSMin
Alpha164	公式正常但逻辑不对	将 Min,函数调整为 TSMin

6 附录

6.1 参考文档

国泰君安：<<**数量化专题：基于短周期价量特征的多因子选股体系**>>

6.2 191 因子表达式

```
Alpha1:   (-1 * CORR(RANK(DELTA(LOG(VOLUME), 1)), RANK(((CLOSE - OPEN) / OPEN)), 6))
Alpha2:   (-1 * DELTA((((CLOSE - LOW) - (HIGH - CLOSE)) / (HIGH - LOW)), 1))
Alpha3:   SUM((CLOSE=DELAY(CLOSE,1)?0:CLOSE-(CLOSE>DELAY(CLOSE,1)?MIN(LOW, DELAY(CLOSE,1)):MAX(HIGH,DELAY(CLOSE,1)))),6)
Alpha4:    ((((SUM(CLOSE, 8) / 8) + STD(CLOSE, 8)) < (SUM(CLOSE, 2) / 2)) ? (-1 * 1) : (((SUM(CLOSE, 2) / 2) <((SUM(CLOSE, 8) / 8) - STD(CLOSE, 8))) ? 1 : (((1 < (VOLUME / MEAN(VOLUME,20))) ||   ((VOLUME /MEAN(VOLUME,20)) == 1)) ? 1 : (-1 * 1))))
Alpha5:   (-1 * TSMAX(CORR(TSRANK(VOLUME, 5), TSRANK(HIGH, 5), 5), 3))
Alpha6:   (RANK(SIGN(DELTA((((OPEN * 0.85) + (HIGH * 0.15))), 4)))* -1)
Alpha7:   ((RANK(MAX((VWAP - CLOSE), 3)) + RANK(MIN((VWAP - CLOSE), 3))) * RANK(DELTA(VOLUME, 3)))
Alpha8:  RANK(DELTA((((HIGH + LOW) / 2) * 0.2) + (VWAP * 0.8)), 4) * -1)
Alpha9:
SMA(((HIGH+LOW)/2-(DELAY(HIGH,1)+DELAY(LOW,1))/2)*(HIGH-LOW)/VOLUME,7,2)
Alpha10: (RANK(MAX(((RET < 0) ? STD(RET, 20) : CLOSE)^2),5))
Alpha11: SUM(((CLOSE-LOW)-(HIGH-CLOSE))./(HIGH-LOW).*VOLUME,6)
Alpha12: (RANK((OPEN - (SUM(VWAP, 10) / 10)))) * (-1 * (RANK(ABS((CLOSE - VWAP)))))
Alpha13: (((HIGH * LOW)^0.5) - VWAP)
Alpha14: CLOSE-DELAY(CLOSE,5)
Alpha15: OPEN/DELAY(CLOSE,1)-1
Alpha16: (-1 * TSMAX(RANK(CORR(RANK(VOLUME), RANK(VWAP), 5)), 5))
Alpha17: RANK((VWAP - MAX(VWAP, 15)))^DELTA(CLOSE, 5)
```

Alpha18: CLOSE/DELAY(CLOSE,5)

Alpha19: (CLOSE<DELAY(CLOSE,5)?(CLOSE-DELAY(CLOSE,5))/DELAY(CLOSE,5):(CLOSE=DELAY(CLOSE,5)?0:(CLOSE-DELAY(CLOSE,5))/CLOSE))

Alpha20: (CLOSE-DELAY(CLOSE,6))/DELAY(CLOSE,6)*100

Alpha21: REGBETA(MEAN(CLOSE,6),SEQUENCE(6))

Alpha22:

SMEAN(((CLOSE-MEAN(CLOSE,6))/MEAN(CLOSE,6)-DELAY((CLOSE-MEAN(CLOSE,6))/MEAN(CLOSE,6),3)),12,1)

Alpha23:

SMA((CLOSE>DELAY(CLOSE,1)?STD(CLOSE:20),0),20,1)/(SMA((CLOSE>DELAY(CLOSE,1)?STD(CLOSE,20):0),20,1)+SMA((CLOSE<=DELAY(CLOSE,1)?STD(CLOSE,20):0),20,1))*100

Alpha24: SMA(CLOSE-DELAY(CLOSE,5),5,1)

Alpha25: ((-1 * RANK((DELTA(CLOSE, 7) * (1 - RANK(DECAYLINEAR((VOLUME /MEAN(VOLUME,20)), 9)))))) * (1 +RANK(SUM(RET, 250))))

Alpha26: ((((SUM(CLOSE, 7) / 7) - CLOSE)) + ((CORR(VWAP, DELAY(CLOSE, 5), 230))))

Alpha27: WMA((CLOSE-DELAY(CLOSE,3))/DELAY(CLOSE,3)*100+(CLOSE-DELAY(CLOSE,6))/DELAY(CLOSE,6)*100,12)

Alpha28: 3*SMA((CLOSE-TSMIN(LOW,9))/(TSMAX(HIGH,9)-TSMIN(LOW,9))*100,3,1)-2*SMA(SMA((CLOSE-TSMIN(LOW,9))/(MAX(HIGH,9)-TSMAX(LOW,9))*100,3,1),3,1)

Alpha29: (CLOSE-DELAY(CLOSE,6))/DELAY(CLOSE,6)*VOLUME

Alpha30: WMA((REGRESI(CLOSE/DELAY(CLOSE)-1,MKT,SMB,HML， 60))^2,20)

Alpha31: (CLOSE-MEAN(CLOSE,12))/MEAN(CLOSE,12)*100

Alpha32: (-1 * SUM(RANK(CORR(RANK(HIGH), RANK(VOLUME), 3)), 3))

Alpha33: ((((-1 * TSMIN(LOW, 5)) + DELAY(TSMIN(LOW, 5), 5)) * RANK(((SUM(RET, 240) - SUM(RET, 20)) / 220))) *TSRANK(VOLUME, 5))Alpha34 MEAN(CLOSE,12)/CLOSE

Alpha35: (MIN(RANK(DECAYLINEAR(DELTA(OPEN, 1), 15)), RANK(DECAYLINEAR(CORR((VOLUME), ((OPEN * 0.65) +(OPEN *0.35)), 17),7))) * -1)

Alpha36: RANK(SUM(CORR(RANK(VOLUME), RANK(VWAP)), 6), 2)

Alpha37: (-1 * RANK(((SUM(OPEN, 5) * SUM(RET, 5)) - DELAY((SUM(OPEN, 5) * SUM(RET, 5)), 10))))

Alpha38: (((SUM(HIGH, 20) / 20) < HIGH) ? (-1 * DELTA(HIGH, 2)) : 0)

Alpha39: ((RANK(DECAYLINEAR(DELTA((CLOSE), 2),8)) - RANK(DECAYLINEAR(CORR(((VWAP * 0.3) + (OPEN * 0.7)),SUM(MEAN(VOLUME,180), 37), 14), 12))) * -1)

Alpha40: SUM((CLOSE>DELAY(CLOSE,1)?VOLUME:0),26)/SUM((CLOSE<=DELAY(CLOSE,1)?VOLUME:0),26)*100

Alpha41: (RANK(MAX(DELTA((VWAP), 3), 5))* -1)

Alpha42: ((-1 * RANK(STD(HIGH, 10))) * CORR(HIGH, VOLUME, 10))

Alpha43: SUM((CLOSE>DELAY(CLOSE,1)?VOLUME:(CLOSE<DELAY(CLOSE,1)?-VOLUME:0)),6)

Alpha44: (TSRANK(DECAYLINEAR(CORR(((LOW)), MEAN(VOLUME,10), 7), 6),4) + TSRANK(DECAYLINEAR(DELTA((VWAP),3), 10), 15))

Alpha45: (RANK(DELTA((((CLOSE * 0.6) + (OPEN *0.4))), 1)) * RANK(CORR(VWAP, MEAN(VOLUME,150), 15)))

Alpha46::

(MEAN(CLOSE,3)+MEAN(CLOSE,6)+MEAN(CLOSE,12)+MEAN(CLOSE,24))/(4*CLOSE)

Alpha47: SMA((TSMAX(HIGH,6)-CLOSE)/(TSMAX(HIGH,6)-TSMIN(LOW,6))*100,9,1)

Alpha48: (-1*((RANK(((SIGN((CLOSE - DELAY(CLOSE, 1))) + SIGN((DELAY(CLOSE, 1) - DELAY(CLOSE, 2)))) +SIGN((DELAY(CLOSE, 2) - DELAY(CLOSE, 3)))))) * SUM(VOLUME, 5)) / SUM(VOLUME, 20))

Alpha49: SUM(((HIGH+LOW)>=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12)/(SUM(((HIGH+LOW)>=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12)+SUM(((HIGH+LOW)<=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12))

Alpha50: SUM(((HIGH+LOW)<=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12)/(SUM(((HIGH+LOW)<=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12)+SUM(((HIGH+LOW)>=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12))-SUM(((HIGH+LOW)>=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12)/(SUM(((HIGH+LOW)>=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12)+SUM(((HIGH+LOW)<=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12))

Alpha51: SUM(((HIGH+LOW)<=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12)/(SUM(((HIGH+LOW)<=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12)+SUM(((HIGH+LOW)>=(DELAY(HIGH,1)+DELAY(LOW,1))?0:MAX(ABS(HIGH-DELAY(HIGH,1)),ABS(LOW-DELAY(LOW,1)))),12))

Alpha52: SUM(MAX(0,HIGH-DELAY((HIGH+LOW+CLOSE)/3,1)),26)/SUM(MAX(0,DELAY((HIGH+LOW+CLOSE)/3,1)-L),26)*100

Alpha53: COUNT(CLOSE>DELAY(CLOSE,1),12)/12*100

Alpha54: (-1 * RANK((STD(ABS(CLOSE - OPEN)) + (CLOSE - OPEN)) + CORR(CLOSE, OPEN,10)))

Alpha55: SUM(16*(CLOSE-DELAY(CLOSE,1)+(CLOSE-OPEN)/2+DELAY(CLOSE,1)-DELAY(OPEN,1))/((ABS(HIGH-DELAY(CLOSE,1))>ABS(LOW-DELAY(CLOSE,1))&ABS(HIGH-DELAY(CLOSE,1))>ABS(HIGH-DELAY(LOW,1))?ABS(HIGH-DELAY(CLOSE,1))+ABS(LOW-DELAY(CLOSE,1))/2+ABS(DELAY(CLOSE,1)-DELAY(OPEN,1))/4:(ABS(LOW-DELAY(CLOSE,1))>ABS(HIGH-DELAY(LOW,1))&ABS(LOW-DELAY(CLOSE,1))>ABS(HIGH-DELAY(CLOSE,1))?ABS(LOW-DELAY(CLOSE,1))+ABS(HIGH-DELAY(CLOSE,1))/2+ABS(DELAY(CLOSE,1)-DELAY(OPEN,1))/4:ABS(HIGH-DELAY(LOW,1))+ABS(DELAY(CLOSE,1)-DELAY(OPEN,1))/4)))*MAX(ABS(HIGH-DELAY(CLOSE,1)),ABS(LOW-DELAY(CLOSE,1))),20)

Alpha56: (RANK((OPEN - TSMIN(OPEN, 12))) < RANK((RANK(CORR(SUM(((HIGH + LOW) / 2), 19),
SUM(MEAN(VOLUME,40), 19), 13))^5)))

Alpha57: SMA((CLOSE-TSMIN(LOW,9))/(TSMAX(HIGH,9)-TSMIN(LOW,9))*100,3,1)

Alpha58: COUNT(CLOSE>DELAY(CLOSE,1),20)/20*100

Alpha59:
SUM((CLOSE=DELAY(CLOSE,1)?0:CLOSE-(CLOSE>DELAY(CLOSE,1)?MIN(LOW,DELAY(CLOSE,1)):MAX(HIGH,DELAY(CLOSE,1)))),20)

Alpha60: SUM(((CLOSE-LOW)-(HIGH-CLOSE))./(HIGH-LOW).*VOLUME,20)

Alpha61: (MAX(RANK(DECAYLINEAR(DELTA(VWAP, 1),
12)),RANK(DECAYLINEAR(RANK(CORR((LOW),MEAN(VOLUME,80), 8)), 17))) * -1)

Alpha62: (-1 * CORR(HIGH, RANK(VOLUME), 5))

Alpha63:
SMA(MAX(CLOSE-DELAY(CLOSE,1),0),6,1)/SMA(ABS(CLOSE-DELAY(CLOSE,1)),6,1)*100

Alpha64: (MAX(RANK(DECAYLINEAR(CORR(RANK(VWAP), RANK(VOLUME), 4),
4)),RANK(DECAYLINEAR(MAX(CORR(RANK(CLOSE), RANK(MEAN(VOLUME,60)), 4), 13), 14)))
* -1)

Alpha65: MEAN(CLOSE,6)/CLOSE

Alpha66: (CLOSE-MEAN(CLOSE,6))/MEAN(CLOSE,6)*100

Alpha67:
SMA(MAX(CLOSE-DELAY(CLOSE,1),0),24,1)/SMA(ABS(CLOSE-DELAY(CLOSE,1)),24,1)*100

Alpha68:
SMA(((HIGH+LOW)/2-(DELAY(HIGH,1)+DELAY(LOW,1))/2)*(HIGH-LOW)/VOLUME,15,2)

Alpha69: (SUM(DTM,20)>SUM(DBM,20)? (SUM(DTM,20)-SUM(DBM,20))/SUM(DTM,20):
(SUM(DTM,20)=SUM(DBM,20)? 0: (SUM(DTM,20)-SUM(DBM,20))/SUM(DBM,20)))

Alpha70: STD(AMOUNT,6)

Alpha71: (CLOSE-MEAN(CLOSE,24))/MEAN(CLOSE,24)*100

Alpha72: SMA((TSMAX(HIGH,6)-CLOSE)/(TSMAX(HIGH,6)-TSMIN(LOW,6))*100,15,1)

Alpha73: ((TSRANK(DECAYLINEAR(DECAYLINEAR(CORR((CLOSE), VOLUME, 10), 16), 4), 5)
-RANK(DECAYLINEAR(CORR(VWAP, MEAN(VOLUME,30), 4),3))) * -1)

Alpha74: (RANK(CORR(SUM(((LOW * 0.35) + (VWAP * 0.65)), 20), SUM(MEAN(VOLUME,40),
20), 7)) +RANK(CORR(RANK(VWAP), RANK(VOLUME), 6)))

Alpha75: COUNT(CLOSE>OPEN
&BANCHMARKINDEXCLOSE<BANCHMARKINDEXOPEN,50)/COUNT(BANCHMARKINDEXCLOSE<BANCHMA
RKINDEXOPEN,50)

Alpha76: STD(ABS((CLOSE/DELAY(CLOSE,1)-1))/VOLUME,20)/MEAN(ABS((CLOSE/DELAY
(CLOSE,1)-1))/VOLUME,20)

Alpha77: MIN(RANK(DECAYLINEAR((((HIGH + LOW) / 2) + HIGH) - (VWAP + HIGH)),
20)),RANK(DECAYLINEAR(CORR(((HIGH + LOW) / 2), MEAN(VOLUME,40), 3), 6)))

Alpha78: ((HIGH+LOW+CLOSE)/3-MA((HIGH+LOW+CLOSE)/3,12))/(0.015*MEAN(ABS(CLOSE-
MEAN((HIGH+LOW+CLOSE)/3,12)),12))

Alpha79:
SMA(MAX(CLOSE-DELAY(CLOSE,1),0),12,1)/SMA(ABS(CLOSE-DELAY(CLOSE,1)),12,1)*100

Alpha80: (VOLUME-DELAY(VOLUME,5))/DELAY(VOLUME,5)*100

Alpha81: SMA(VOLUME,21,2)

Alpha82: SMA((TSMAX(HIGH,6)-CLOSE)/(TSMAX(HIGH,6)-TSMIN(LOW,6))*100,20,1)

Alpha83: (-1 * RANK(COVIANCE(RANK(HIGH), RANK(VOLUME), 5)))

Alpha84:
SUM((CLOSE>DELAY(CLOSE,1)?VOLUME:(CLOSE<DELAY(CLOSE,1)?-VOLUME:0)),20)

Alpha85: (TSRANK((VOLUME / MEAN(VOLUME,20)), 20) * TSRANK((-1 * DELTA(CLOSE, 7)),
8))

Alpha86: ((0.25 < (((DELAY(CLOSE, 20) - DELAY(CLOSE, 10)) / 10) - ((DELAY(CLOSE, 10) - CLOSE) / 10))) ? (-1 * 1) :(((((DELAY(CLOSE, 20) - DELAY(CLOSE, 10)) / 10) - ((DELAY(CLOSE, 10) - CLOSE) / 10)) < 0) ? 1 : ((-1 * 1) *(CLOSE - DELAY(CLOSE, 1)))))

Alpha87: ((RANK(DECAYLINEAR(DELTA(VWAP, 4), 7)) + TSRANK(DECAYLINEAR((((LOW * 0.9) + (LOW * 0.1)) - VWAP) /(OPEN - ((HIGH + LOW) / 2))), 11), 7)) * -1)

Alpha88: (CLOSE-DELAY(CLOSE,20))/DELAY(CLOSE,20)*100

Alpha89:
2*(SMA(CLOSE,13,2)-SMA(CLOSE,27,2)-SMA(SMA(CLOSE,13,2)-SMA(CLOSE,27,2),10,2))

Alpha90: (RANK(CORR(RANK(VWAP), RANK(VOLUME), 5)) * -1)

Alpha91: ((RANK((CLOSE - MAX(CLOSE, 5)))*RANK(CORR((MEAN(VOLUME,40)), LOW, 5))) * -1)

Alpha92: (MAX(RANK(DECAYLINEAR(DELTA(((CLOSE * 0.35) + (VWAP *0.65)), 2), 3)),TSRANK(DECAYLINEAR(ABS(CORR((MEAN(VOLUME,180)), CLOSE, 13)), 5), 15)) * -1)

Alpha93:
SUM((OPEN>=DELAY(OPEN,1)?0:MAX((OPEN-LOW),(OPEN-DELAY(OPEN,1)))),20)

Alpha94:
SUM((CLOSE>DELAY(CLOSE,1)?VOLUME:(CLOSE<DELAY(CLOSE,1)?-VOLUME:0)),30)

Alpha95: STD(AMOUNT,20)

Alpha96:
SMA(SMA((CLOSE-TSMIN(LOW,9))/(TSMAX(HIGH,9)-TSMIN(LOW,9))*100,3,1),3,1)

Alpha97: STD(VOLUME,10)

Alpha98: ((((DELTA((SUM(CLOSE, 100) / 100), 100) / DELAY(CLOSE, 100)) < 0.05) || ((DELTA((SUM(CLOSE, 100) / 100), 100) /DELAY(CLOSE, 100)) == 0.05)) ? (-1 * (CLOSE - TSMIN(CLOSE, 100))) : (-1 * DELTA(CLOSE, 3)))

Alpha99: (-1 * RANK(COVIANCE(RANK(CLOSE), RANK(VOLUME), 5)))

Alpha100: STD(VOLUME,20)

Alpha101: ((RANK(CORR(CLOSE, SUM(MEAN(VOLUME,30), 37), 15)) < RANK(CORR(RANK(((HIGH * 0.1) + (VWAP * 0.9))),RANK(VOLUME), 11))) * -1)

Alpha102: SMA(MAX(VOLUME-DELAY(VOLUME,1),0),6,1)/SMA(ABS(VOLUME-DELAY(VOLUME,1)),6,1)*100

Alpha103: ((20-LOWDAY(LOW,20))/20)*100

Alpha104: (-1 * (DELTA(CORR(HIGH, VOLUME, 5), 5) * RANK(STD(CLOSE, 20))))

Alpha105: (-1 * CORR(RANK(OPEN), RANK(VOLUME), 10))

Alpha106: CLOSE-DELAY(CLOSE,20)

Alpha107: (((-1 * RANK((OPEN - DELAY(HIGH, 1)))) * RANK((OPEN - DELAY(CLOSE, 1)))) *RANK((OPEN - DELAY(LOW, 1))))

Alpha108: ((RANK((HIGH - MIN(HIGH, 2)))^RANK(CORR((VWAP), (MEAN(VOLUME,120)), 6))) * -1)

Alpha109: SMA(HIGH-LOW,10,2)/SMA(SMA(HIGH-LOW,10,2),10,2)

Alpha110:
SUM(MAX(0,HIGH-DELAY(CLOSE,1)),20)/SUM(MAX(0,DELAY(CLOSE,1)-LOW),20)*100

Alpha111:
SMA(VOL*((CLOSE-LOW)-(HIGH-CLOSE))/(HIGH-LOW),11,2)-SMA(VOL*((CLOSE-

LOW)-(HIGH-CLOSE))/(HIGH-LOW),4,2)

Alpha11:
(SUM((CLOSE-DELAY(CLOSE,1)>0?CLOSE-DELAY(CLOSE,1):0),12)-SUM((CLOSE-DELAY(CLOSE,1)<0?ABS(CLOSE-DELAY(CLOSE,1)):0),12))/(SUM((CLOSE-DELAY(CLOSE,1)>0?CLOSE-DELAY(CLOSE,1):0),12)+SUM((CLOSE-DELAY(CLOSE,1)<0?ABS(CLOSE-DELAY(CLOSE,1)):0),12))*100

Alpha113: (-1 * ((RANK((SUM(DELAY(CLOSE, 5), 20) / 20)) * CORR(CLOSE, VOLUME, 2)) *RANK(CORR(SUM(CLOSE, 5),SUM(CLOSE, 20), 2))))

Alpha114: ((RANK(DELAY(((HIGH - LOW) / (SUM(CLOSE, 5) / 5)), 2)) * RANK(RANK(VOLUME))) / (((HIGH - LOW) /(SUM(CLOSE, 5) / 5)) / (VWAP - CLOSE)))

Alpha115: (RANK(CORR(((HIGH * 0.9) + (CLOSE * 0.1)), MEAN(VOLUME,30), 10))^RANK(CORR(TSRANK(((HIGH + LOW) /2), 4), TSRANK(VOLUME, 10), 7)))

Alpha116: REGBETA(CLOSE,SEQUENCE,20)

Alpha117: ((TSRANK(VOLUME, 32) * (1 - TSRANK(((CLOSE + HIGH) - LOW), 16))) * (1 - TSRANK(RET, 32)))

Alpha118: SUM(HIGH-OPEN,20)/SUM(OPEN-LOW,20)*100

Alpha119: (RANK(DECAYLINEAR(CORR(VWAP, SUM(MEAN(VOLUME,5), 26), 5), 7)) -RANK(DECAYLINEAR(TSRANK(MIN(CORR(RANK(OPEN), RANK(MEAN(VOLUME,15)), 21), 9), 7), 8)))

Alpha120: (RANK((VWAP - CLOSE)) / RANK((VWAP + CLOSE)))

Alpha121: ((RANK((VWAP - MIN(VWAP, 12)))^TSRANK(CORR(TSRANK(VWAP, 20), TSRANK(MEAN(VOLUME,60), 2), 18), 3)) *-1)

Alpha122:
(SMA(SMA(SMA(LOG(CLOSE),13,2),13,2),13,2)-DELAY(SMA(SMA(SMA(LOG(CLOSE),13,2),13,2),13,2),1))/DELAY(SMA(SMA(SMA(LOG(CLOSE),13,2),13,2),13,2),1)

Alpha123: ((RANK(CORR(SUM(((HIGH + LOW) / 2), 20), SUM(MEAN(VOLUME,60), 20), 9)) <RANK(CORR(LOW, VOLUME,6))) * -1)

Alpha124: (CLOSE - VWAP) / DECAYLINEAR(RANK(TSMAX(CLOSE, 30)),2)

Alpha125: (RANK(DECAYLINEAR(CORR((VWAP), MEAN(VOLUME,80),17), 20)) / RANK(DECAYLINEAR(DELTA(((CLOSE * 0.5)+ (VWAP * 0.5)), 3), 16)))

Alpha126: (CLOSE+HIGH+LOW)/3

Alpha127: (MEAN((100*(CLOSE-MAX(CLOSE,12))/(MAX(CLOSE,12)))^2))^(1/2)

Alpha128:
100-(100/(1+SUM(((HIGH+LOW+CLOSE)/3>DELAY((HIGH+LOW+CLOSE)/3,1)?(HIGH+LOW+CLOSE)/3*VOLUME:0),14)/SUM(((HIGH+LOW+CLOSE)/3<DELAY((HIGH+LOW+CLOSE)/3,1)?(HIGH+LOW+CLOSE)/3*VOLUME:0),14)))

Alpha129: SUM((CLOSE-DELAY(CLOSE,1)<0?ABS(CLOSE-DELAY(CLOSE,1)):0),12)

Alpha130: (RANK(DECAYLINEAR(CORR(((HIGH + LOW) / 2), MEAN(VOLUME,40), 9), 10)) /RANK(DECAYLINEAR(CORR(RANK(VWAP), RANK(VOLUME), 7),3)))

Alpha131:(RANK(DELAT(VWAP, 1))^TSRANK(CORR(CLOSE,MEAN(VOLUME,50), 18), 18))

Alpha132: MEAN(AMOUNT,20)

Alpha133: ((20-HIGHDAY(HIGH,20))/20)*100-((20-LOWDAY(LOW,20))/20)*100

Alpha134: (CLOSE-DELAY(CLOSE,12))/DELAY(CLOSE,12)*VOLUME

Alpha135: SMA(DELAY(CLOSE/DELAY(CLOSE,20),1),20,1)

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Alpha136: ((-1 * RANK(DELTA(RET, 3))) * CORR(OPEN, VOLUME, 10))
Alpha137:
16*(CLOSE-DELAY(CLOSE,1)+(CLOSE-OPEN)/2+DELAY(CLOSE,1)-DELAY(OPEN,1))/
((ABS(HIGH-DELAY(CLOSE,1))>ABS(LOW-DELAY(CLOSE,1))
&ABS(HIGH-DELAY(CLOSE,1))>ABS(HIGH-DELAY(LOW,1))?ABS(HIGH-DELAY(CLOSE,1))
+ABS(LOW-DELAY(CLOSE,1))/2+ABS(DELAY(CLOSE,1)-DELAY(OPEN,1))/4:(ABS(LOW-D
ELAY(CLOSE,1))>ABS(HIGH-DELAY(LOW,1))
&ABS(LOW-DELAY(CLOSE,1))>ABS(HIGH-DELAY(CLOSE,1))?ABS(LOW-DELAY(CLOSE,1))
+ABS(HIGH-DELAY(CLOSE,1))/2+ABS(DELAY(CLOSE,1)-DELAY(OPEN,1))/4:ABS(HIGH-
DELAY(LOW,1))+ABS(DELAY(CLOSE,1)-DELAY(OPEN,1))/4)))*MAX(ABS(HIGH-DELAY(C
LOSE,1)),ABS(LOW-DELAY(CLOSE,1)))
Alpha138: ((RANK(DECAYLINEAR(DELTA((((LOW * 0.7) + (VWAP *0.3))), 3), 20))
-TSRANK(DECAYLINEAR(TSRANK(CORR(TSRANK(LOW, 8), TSRANK(MEAN(VOLUME,60),
17), 5), 19), 16), 7)) * -1)
Alpha139: (-1 * CORR(OPEN, VOLUME, 10))
Alpha140: MIN(RANK(DECAYLINEAR(((RANK(OPEN) + RANK(LOW)) - (RANK(HIGH) +
RANK(CLOSE))), 8)),TSRANK(DECAYLINEAR(CORR(TSRANK(CLOSE, 8),
TSRANK(MEAN(VOLUME,60), 20), 8), 7), 3))
Alpha141: (RANK(CORR(RANK(HIGH), RANK(MEAN(VOLUME,15)), 9))* -1)
Alpha142: (((-1 * RANK(TSRANK(CLOSE, 10))) * RANK(DELTA(DELTA(CLOSE, 1), 1)))
*RANK(TSRANK((VOLUME/MEAN(VOLUME,20)), 5)))
Alpha143:
CLOSE>DELAY(CLOSE,1)?(CLOSE-DELAY(CLOSE,1))/DELAY(CLOSE,1)*SELF:SELF
Alpha144:
SUMIF(ABS(CLOSE/DELAY(CLOSE,1)-1)/AMOUNT,20,CLOSE<DELAY(CLOSE,1))/
COUNT(CLOSE<DELAY(CLOSE,1),20)
Alpha145: (MEAN(VOLUME,9)-MEAN(VOLUME,26))/MEAN(VOLUME,12)*100
Alpha146:
MEAN((CLOSE-DELAY(CLOSE,1))/DELAY(CLOSE,1)-SMA((CLOSE-DELAY(CLOSE,1))/
DELAY(CLOSE,1),61,2),20)*((CLOSE-DELAY(CLOSE,1))/DELAY(CLOSE,1)-SMA((CLOS
E-DELAY(CLOSE,1))/DELAY(CLOSE,1),61,2))/SMA(((CLOSE-DELAY(CLOSE,1))/DELAY
(CLOSE,1)-((CLOSE-DELAY(CLOSE,1))/DELAY(CLOSE,1)-SMA((CLOSE-DELAY(CLOSE,1
))/DELAY(CLOSE,1),61,2)))^2,60);
Alpha147: REGBETA(MEAN(CLOSE,12),SEQUENCE(12))
Alpha148: ((RANK(CORR((OPEN), SUM(MEAN(VOLUME,60), 9), 6)) < RANK((OPEN -
TSMIN(OPEN, 14)))) * -1)
Alpha14: REGBETA(FILTER(CLOSE/DELAY(CLOSE,1)-1,BANCHMARKINDEXCLOSE<
DELAY(BANCHMARKINDEXCLOSE,1)),FILTER(BANCHMARKINDEXCLOSE/DELAY(BANCHMARKI
NDEXCLOSE,1)-1,BANCHMARKINDEXCLOSE<DELAY(BANCHMARKINDEXCLOSE,1)),252)
Alpha150: (CLOSE+HIGH+LOW)/3*VOLUME
Alpha151: SMA(CLOSE-DELAY(CLOSE,20),20,1)
Alpha152: SMA(MEAN(DELAY(SMA(DELAY(CLOSE/DELAY(CLOSE,9),1),9,1),1),12)-
MEAN(DELAY(SMA(DELAY(CLOSE/DELAY(CLOSE,9),1),9,1),1),26),9,1)
Alpha153: (MEAN(CLOSE,3)+MEAN(CLOSE,6)+MEAN(CLOSE,12)+MEAN(CLOSE,24))/4
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Alpha154: (((VWAP - MIN(VWAP, 16))) < (CORR(VWAP, MEAN(VOLUME,180), 18)))

Alpha155:
SMA(VOLUME,13,2)-SMA(VOLUME,27,2)-SMA(SMA(VOLUME,13,2)-SMA(VOLUME,27,2),10,2)

Alpha156: (MAX(RANK(DECAYLINEAR(DELTA(VWAP, 5), 3)),RANK(DECAYLINEAR(((DELTA(((OPEN * 0.15) + (LOW *0.85)),2) / ((OPEN * 0.15) + (LOW * 0.85))) * -1), 3))) * -1)

Alpha157: (MIN(PROD(RANK(RANK(LOG(SUM(TSMIN(RANK(RANK((-1 * RANK(DELTA((CLOSE - 1), 5))))), 2), 1)))), 1), 5) +TSRANK(DELAY((-1 * RET), 6), 5))

Alpha158: ((HIGH-SMA(CLOSE,15,2))-(LOW-SMA(CLOSE,15,2)))/CLOSE

Alpha159:
((CLOSE-SUM(MIN(LOW,DELAY(CLOSE,1)),6))/SUM(MAX(HGIH,DELAY(CLOSE,1))-MIN(LOW,DELAY(CLOSE,1)),6)*12*24+(CLOSE-SUM(MIN(LOW,DELAY(CLOSE,1)),12))/SUM(MAX(HGIH,DELAY(CLOSE,1))-MIN(LOW,DELAY(CLOSE,1)),12)*6*24+(CLOSE-SUM(MIN(LOW,DELAY(CLOSE,1)),24))/SUM(MAX(HGIH,DELAY(CLOSE,1))-MIN(LOW,DELAY(CLOSE,1)),24)*6*24)*100/(6*12+6*24+12*24)

Alpha160: SMA((CLOSE<=DELAY(CLOSE,1)?STD(CLOSE,20):0),20,1)

Alpha16:
MEAN(MAX(MAX((HIGH-LOW),ABS(DELAY(CLOSE,1)-HIGH)),ABS(DELAY(CLOSE,1)-LOW)),12)

Alpha162:
(SMA(MAX(CLOSE-DELAY(CLOSE,1),0),12,1)/SMA(ABS(CLOSE-DELAY(CLOSE,1)),12,1)
*100-MIN(SMA(MAX(CLOSE-DELAY(CLOSE,1),0),12,1)/SMA(ABS(CLOSE-DELAY(CLOSE,1)),12,1)*100,12))/(MAX(SMA(MAX(CLOSE-DELAY(CLOSE,1),0),12,1)/SMA(ABS(CLOSE-DELAY(CLOSE,1)),12,1)*100,12)-MIN(SMA(MAX(CLOSE-DELAY(CLOSE,1),0),12,1)/SMA(ABS(CLOSE-DELAY(CLOSE,1)),12,1)*100,12))

Alpha163: RANK(((((-1 * RET) * MEAN(VOLUME,20)) * VWAP) * (HIGH - CLOSE)))

Alpha164:
SMA((((CLOSE>DELAY(CLOSE,1))?1/(CLOSE-DELAY(CLOSE,1)):1)-MIN(((CLOSE>DELAY(CLOSE,1))?1/(CLOSE-DELAY(CLOSE,1)):1),12))/(HIGH-LOW)*100,13,2)

Alpha165:
MAX(SUMAC(CLOSE-MEAN(CLOSE,48)))-MIN(SUMAC(CLOSE-MEAN(CLOSE,48)))/STD(CLOSE,48)

Alpha166: -20*（ 20-1 ）
^1.5*SUM(CLOSE/DELAY(CLOSE,1)-1-MEAN(CLOSE/DELAY(CLOSE,1)-1,20),20)/((20-1)*(20-2)(SUM((CLOSE/DELAY(CLOSE,1),20)^2,20))^1.5)

Alpha167: SUM((CLOSE-DELAY(CLOSE,1)>0?CLOSE-DELAY(CLOSE,1):0),12)

Alpha168: (-1*VOLUME/MEAN(VOLUME,20))

Alpha169: SMA(MEAN(DELAY(SMA(CLOSE-DELAY(CLOSE,1),9,1),1),12)-MEAN(DELAY(SMA(CLOSE-DELAY(CLOSE,1),9,1),1),26),10,1)

Alpha170: ((((RANK((1 / CLOSE)) * VOLUME) / MEAN(VOLUME,20)) * ((HIGH * RANK((HIGH - CLOSE))) / (SUM(HIGH, 5) /5))) - RANK((VWAP - DELAY(VWAP, 5))))

Alpha171: ((-1 * ((LOW - CLOSE) * (OPEN^5))) / ((CLOSE - HIGH) * (CLOSE^5)))

Alpha172: MEAN(ABS(SUM((LD>0 & LD>HD)?LD:0,14)*100/SUM(TR,14)-SUM((HD>0 &HD>LD)?HD:0,14)*100/SUM(TR,14))/(SUM((LD>0 & LD>HD)?LD:0,14)*100/SUM(TR,14)+SUM((HD>0 &HD>LD)?HD:0,14)*100/SUM(TR,14))*100,6)

Alpha173: 3*SMA(CLOSE,13,2)-2*SMA(SMA(CLOSE,13,2),13,2)+SMA(SMA(SMA(LOG(CLOSE),13,2),13,2),13,2);

Alpha174: SMA((CLOSE>DELAY(CLOSE,1)?STD(CLOSE,20):0),20,1)

Alpha175: MEAN(MAX(MAX((HIGH-LOW),ABS(DELAY(CLOSE,1)-HIGH)),ABS(DELAY(CLOSE,1)-LOW)),6)

Alpha176: CORR(RANK(((CLOSE - TSMIN(LOW, 12)) / (TSMAX(HIGH, 12) - TSMIN(LOW,12)))),RANK(VOLUME), 6)

Alpha177: ((20-HIGHDAY(HIGH,20))/20)*100

Alpha178: (CLOSE-DELAY(CLOSE,1))/DELAY(CLOSE,1)*VOLUME

Alpha179: (RANK(CORR(VWAP, VOLUME, 4)) *RANK(CORR(RANK(LOW),RANK(MEAN(VOLUME,50)), 12)))

Alpha180: ((MEAN(VOLUME,20) < VOLUME) ? ((-1 * TSRANK(ABS(DELTA(CLOSE, 7)), 60)) *SIGN(DELTA(CLOSE, 7)) : (-1 *VOLUME)))

Alpha181: SUM(((CLOSE/DELAY(CLOSE,1)-1)-MEAN((CLOSE/DELAY(CLOSE,1)-1),20))-(BANCHMARKINDEXCLOSE-MEAN(BANCHMARKINDEXCLOSE,20))^2,20)/SUM((BANCHMARKINDEXCLOSE-MEAN(BANCHMARKINDEXCLOSE,20))^3)

Alpha182: COUNT((CLOSE>OPEN & BANCHMARKINDEXCLOSE>BANCHMARKINDEXOPEN)OR(CLOSE<OPEN &BANCHMARKINDEXCLOSE<BANCHMARKINDEXOPEN),20)/20

Alpha183: MAX(SUMAC(CLOSE-MEAN(CLOSE,24)))-MIN(SUMAC(CLOSE-MEAN(CLOSE,24)))/STD(CLOSE,24)

Alpha184: (RANK(CORR(DELAY((OPEN - CLOSE), 1), CLOSE, 200)) + RANK((OPEN - CLOSE)))

Alpha185: RANK((-1 * ((1 - (OPEN / CLOSE))^2)))

Alpha186: (MEAN(ABS(SUM((LD>0 & LD>HD)?LD:0,14)*100/SUM(TR,14)-SUM((HD>0 &HD>LD)?HD:0,14)*100/SUM(TR,14))/(SUM((LD>0 &LD>HD)?LD:0,14)*100/SUM(TR,14)+SUM((HD>0 &HD>LD)?HD:0,14)*100/SUM(TR,14))*100,6)+DELAY(MEAN(ABS(SUM((LD>0 &LD>HD)?LD:0,14)*100/SUM(TR,14)-SUM((HD>0 & HD>LD)?HD:0,14)*100/SUM(TR,14))/(SUM((LD>0 &LD>HD)?LD:0,14)*100/SUM(TR,14)+SUM((HD>0 & HD>LD)?HD:0,14)*100/SUM(TR,14))*100,6),6))/2

Alpha187: SUM((OPEN<=DELAY(OPEN,1)?0:MAX((HIGH-OPEN),(OPEN-DELAY(OPEN,1)))),20)

Alpha188: ((HIGH-LOW-SMA(HIGH-LOW,11,2))/SMA(HIGH-LOW,11,2))*100

Alpha189: MEAN(ABS(CLOSE-MEAN(CLOSE,6)),6)

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Alpha190：
LOG((COUNT(CLOSE/DELAY(CLOSE)-1>((CLOSE/DELAY(CLOSE,19))^(1/20)-1),20)-1)
*
(SUMIF(((CLOSE/DELAY(CLOSE)-1-(CLOSE/DELAY(CLOSE,19))^(1/20)-1))^2,20,CLO
SE/DELAY(CLOSE)-1<(CLOSE/DELAY(CLOSE,19))^(1/20)-1))/((COUNT((CLOSE/DELAY
(CLOSE)-1<(CLOSE/DELAY(CLOSE,19))^(1/20)-1),20))*(SUMIF((CLOSE/DELAY(CLOS
E)-1-((CLOSE/DELAY(CLOSE,19))^(1/20)-1))^2,20,CLOSE/DELAY(CLOSE)-1>(CLOSE
/DELAY(CLOSE,19))^(1/20)-1))))
Alpha191: ((CORR(MEAN(VOLUME,20), LOW, 5) + ((HIGH + LOW) / 2)) - CLOSE)
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